



Eng. Abdelrahman Mahmoud
— DATA ENGINEER —

DATA ANALYSIS
BOOTCAMP

FIFA WORLD CUP ANALYTICS



PLAYERS
TEAMS
MATCHES
TOURNAMENTS



REAL DATA
REAL INSIGHTS
BIGGER STORIES



FIFA
WORLD CUP



FROM DATA
TO VICTORY

Data 
Strikers

FIFA WORLD CUP DATA INTELLIGENCE

Database, Analytics &
Interactive Visualization System



FIFA WORLD CUP DATA INTELLIGENCE

Database, Analytics & Interactive Visualization System

University-level software project documentation for an integrated FIFA World Cup relational database, analytical SQL portfolio, Python analysis layer, statistical checks, and interactive website presentation.

Document scope: concise technical and analytical documentation of the FIFA World Cup Analytics project, covering its data sources, database design, validation, analytics, Python workflow, and project artifacts.

Project information	Entry
Student Name	Aya Khaled — 2301888 Shimaa Nashat — 2401668 Razan Ali — 2301407 Roxana Emad — 2500865 Moreen Romany — 2500831
Project	FIFA World Cup Analytics
Instructor	Abdelrahman Mahmoud
Identified technologies	Microsoft SQL Server, SQL, Python, Pandas, NumPy, Matplotlib, Seaborn, NetworkX, pyodbc, SciPy, and an interactive website layer

Table of Contents

The numbered headings below are formatted as document headings so an automatic table of contents with page numbers can be generated where supported.

Section	Title	Page
1	Introduction	3
2	Project Overview & Objectives	3
3	Data Sources & Dataset	4
4	Database Design	5
5	Data Cleaning & Validation	5
6	Analytics & Insights	6
7	Technology Stack & System Architecture	7
8	Conclusion & Future Improvements	9
9	References / Project Artifacts	9

1. Introduction

1.1 Background and problem statement

FIFA World Cup information is distributed across historical tournament records, team and player entities, match events, stadium information, rankings, attendance, and detailed performance data. Treating these domains as isolated files limits analysis and makes it difficult to validate relationships, compare eras, or connect match outcomes with player and team performance.

This project addresses that problem by integrating the available data into a relational database and connecting database analysis with Python-based exploration, statistical checks, visualizations, and an interactive website.

1.2 Purpose and value

The purpose is to create a reproducible analytics system that supports evidence-based questions about tournaments, teams, players, matches, stadiums, rankings, attendance, and detailed 2026 performance. The project distinguishes source facts, derived measures, statistical associations, and data-quality limitations.

1.3 Objectives

- Collect and integrate World Cup data from the five documented source categories.
- Clean, enrich, validate, and load the data into Microsoft SQL Server.
- Implement a normalized relational design with keys and integrity constraints.
- Develop 32 meaningful analytical SQL queries and connect SQL Server to Python.
- Produce statistical checks, 11 primary visualizations, 14 cross-query insights, and an interactive website presentation.

2. Project Overview & Objectives

2.1 End-to-end system

The system follows the sequence **Data Sources → Data Collection → Data Cleaning → Data Integration → SQL Server Database → SQL Analytics → Python/Pandas/Statistics → Visualization → Interactive Website**. The main scope is the FIFA Men's World Cup, with detailed performance questions using the available detailed 2026 Men's World Cup data.

2.2 Users and purpose

The intended users are project assessors, students, analysts, and readers who need to inspect the database structure, understand analytical decisions, review validated results, and explore the project through the website presentation layer.

2.3 Main features

Capability	Contribution
Relational storage	Connects 17 related football data tables through keys and association tables.
Data quality management	Preserves missing values, applies supported enrichment, records corrections, and validates integrity.
Analytical SQL	Provides 32 queries covering foundational, join, aggregate, ranking, CTE, and view-based analysis.
Python analysis	Uses SQL Server outputs in Pandas for EDA, statistical checks, and selected visualizations.
Interactive presentation	Presents the database, analysis, insights, and visualizations through a website layer.

Design principle. The project is an end-to-end analytics pipeline rather than a simple database exercise: source coverage and analytical limitations remain visible instead of being hidden by fabricated completeness.

3. Data Sources & Dataset

3.1 Documented source categories

Source category	Documented use
Fjelstul World Cup Database / DataHub	Historical teams, players, matches, goals, bookings, substitutions, stadiums, and tournaments.
Detailed World Cup 2026 / FIFA Training Centre Data	Player appearances, player-match statistics, team-match statistics, passing-network edges, and match-performance metrics.
FIFA World Ranking Data	Historical ranking observations and comparison with tournament performance.
FIFA Official 2026 Stadium Information	Completion of missing 2026 stadium city, country, and capacity information.

dim_players.csv	Machine-readable 2026 player birth-date and related metadata enrichment.
-----------------	--

3.2 Dataset coverage

The final database contains **17 tables and 211,808 records**. Its domains are confederations, federations, teams, players, tournaments, stadiums, matches, goals, bookings, substitutions, attendance, player and team appearances, player and team match statistics, passing-network edges, and FIFA rankings.

Important entity	Records
Players	11,364
Tournaments	31
Stadiums	256
Matches	1,352
Player_Appeances	32,824
Goals	3,945
Match_Attendance	964
Passing_Network_Edges	52,072
FIFA_Rankings	71,019
Total database records	211,808

Integration principles: preserve source identities where possible; connect domains with foreign keys; preserve missing information; add only evidence-based enrichment; correct identified issues explicitly; and keep the final state reproducible through the named project artifacts.

Coverage limitations: attendance is available for 964 of 1,352 matches; detailed 2026 metrics do not necessarily have identical coverage; missing line-up or appearance data indicates source coverage limitations rather than non-participation.

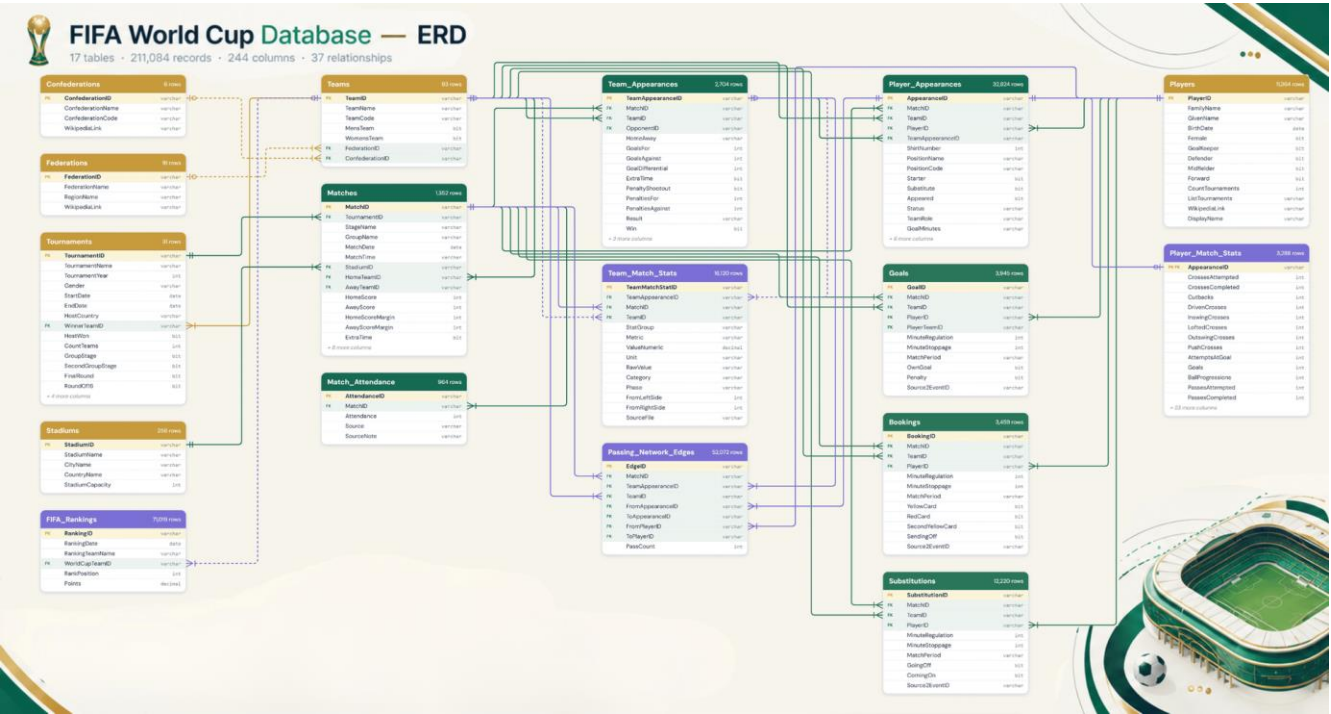
4. Database Design

4.1 Relational architecture

The database is implemented in **Microsoft SQL Server**, named **FIFA_World_Cup_Analytics**, using the default **dbo** schema. The design is normalized and uses primary keys, foreign keys, NOT NULL constraints, UNIQUE constraints, CHECK constraints, association tables, source-oriented attributes, and optional relationships where coverage is incomplete.

Main entities and roles	Examples
Reference entities	Confederations, Federations, Teams, Players.
Competition entities	Tournaments, Stadiums, Matches.
Participation entities	Team_Appearances, Player_Appearances.
Event entities	Goals, Bookings, Substitutions, Match_Attendance.
Performance entities	Player_Match_Stats, Team_Match_Stats, Passing_Network_Edges.
External indicator	FIFA_Rankings, with optional WorldCupTeamID linkage.

4.2 Relationships and cardinalities



The project documentation records **37 relationships**. The supplied ERD reference, **FIFA_WorldCup_ERD.png** visually confirms the documented **17-table relational schema** and its connected reference, competition, participation, event, performance, and ranking entities.

Core cardinalities are Confederations 1:N Teams; Federations 1:N Teams; Tournaments 1:N Matches; and Stadiums 1:N Matches. Matches reference home and away teams as two roles, and connect to goals, bookings, substitutions, attendance, team appearances, and player appearances.

Team appearances connect matches and teams. Player appearances connect matches, players, and the correct team-match context. Player_Match_Stats is a one-to-one extension of a player appearance. Team_Match_Stats extends team performance records; Passing_Network_Edges connects a match, team appearance, team, source player appearance, destination player appearance, and both player entities. FIFA_Rankings.WorldCupTeamID is an optional foreign key to Teams.

Figure 1. FIFA World Cup database ERD. The supplied ERD reference shows the project's 17-table relational schema, including primary-key and foreign-key connections across teams, tournaments, matches, appearances, events, statistics, attendance, passing networks, and rankings. In this responsive document, the image is documented as a reference rather than embedded directly.

5. Data Cleaning & Validation

5.1 Cleaning philosophy and implemented logic

The project preserves missing values rather than filling them for appearance. No synthetic player or match records were created, attendance was not fabricated, missing attendance was not converted to zero, records were not deleted only because attributes were missing, and unsupported birth dates were not invented.

Supported 2026 player metadata was enriched; 16 stadium records were completed using official FIFA information. Documented corrections include M047, M088, M099/M100/M104, and the Crysencio duplicate. The optional Ghana duplicate deletion remains inactive. Updates in 03_update_missing_data.sql are protected by transaction handling.

5.2 Validation results

Check	Result
All 17 table counts matched	PASS
CSV total and SQL Server total	211,808 and 211,808; difference 0
Two team appearances per match	PASS
Every match has team appearances	PASS
Player statistics to appearances	PASS
Passing-network relationship IDs	PASS
Updated 2026 players	963

Players still missing birth date	0
Players still missing position	0
Team_Match_Stats missing TeamAppearanceID and TeamID	104 rows; documented source-coverage warning, not a broken relationship
Attendance/capacity validation	99 records flagged for review; treated as data-quality or historical-definition warnings, not confirmed over-capacity attendance

Validation coverage included: missing values, duplicates, data types, date ordering and ranges, referential integrity, uniqueness, attendance/capacity context, and source-coverage warnings.

6. Analytics & Insights

6.1 Analytical portfolio

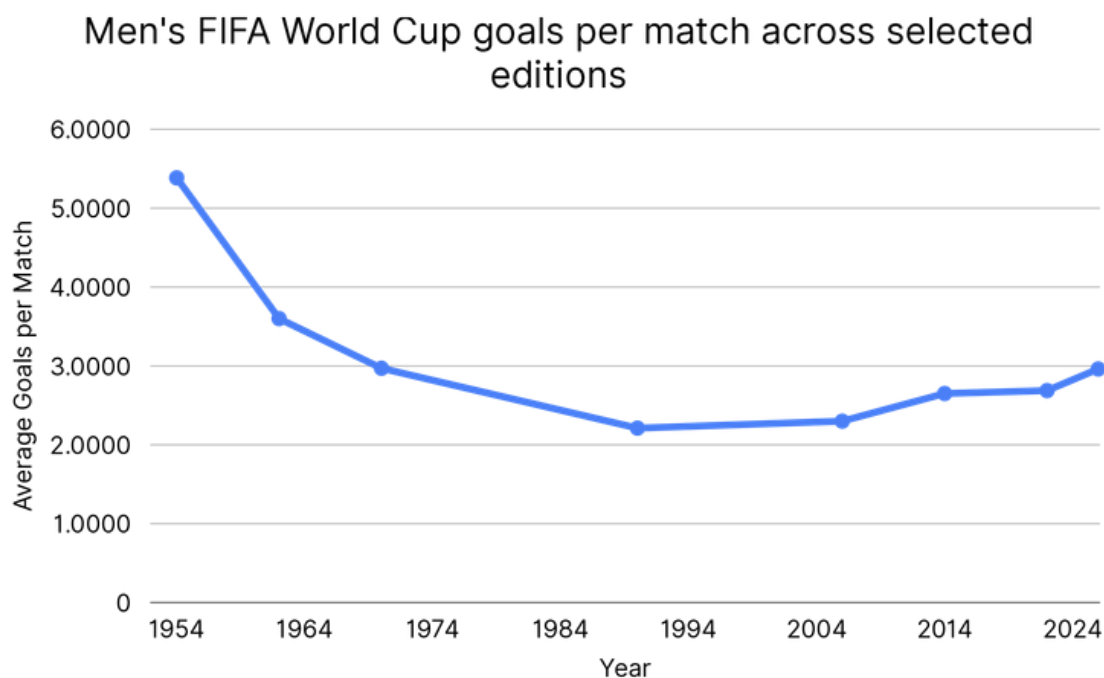
The project contains **32 meaningful analytical SQL queries** covering stadiums, matches, attendance, scoring, players, teams, passing connections, detailed match statistics, rankings, CTEs, window functions, and a reusable team-performance view. The final presentation selects **11 primary visuals** rather than displaying all 32 queries as charts.

Measure	Supported result
Top scorer	Kylian Mbappé – 22 recorded non-own goals.
Most active player	Lionel Messi – 34 recorded appearances.
Top passing edge	Aymeric Laporte → Rodri – 138 passes.
Historical win-rate examples	Brazil 68.91%; Germany 64.81%; West Germany 62.90%; Argentina 62.50%.
Team xG-minus-actual examples	Ecuador +0.97; Netherlands -1.50.
Ranking association	48 matched observations; Spearman rho -0.7531; p = 6.600889727e-10.

6.2 Selected findings and visual evidence

Average goals per match varies by era: 1954 recorded 5.3846, 1990 recorded 2.2115, 2022 recorded 2.6875, and 2026 recorded 2.9615. The 2026 value is a supplied tournament measure and should be interpreted alongside changing tournament formats and exposure.

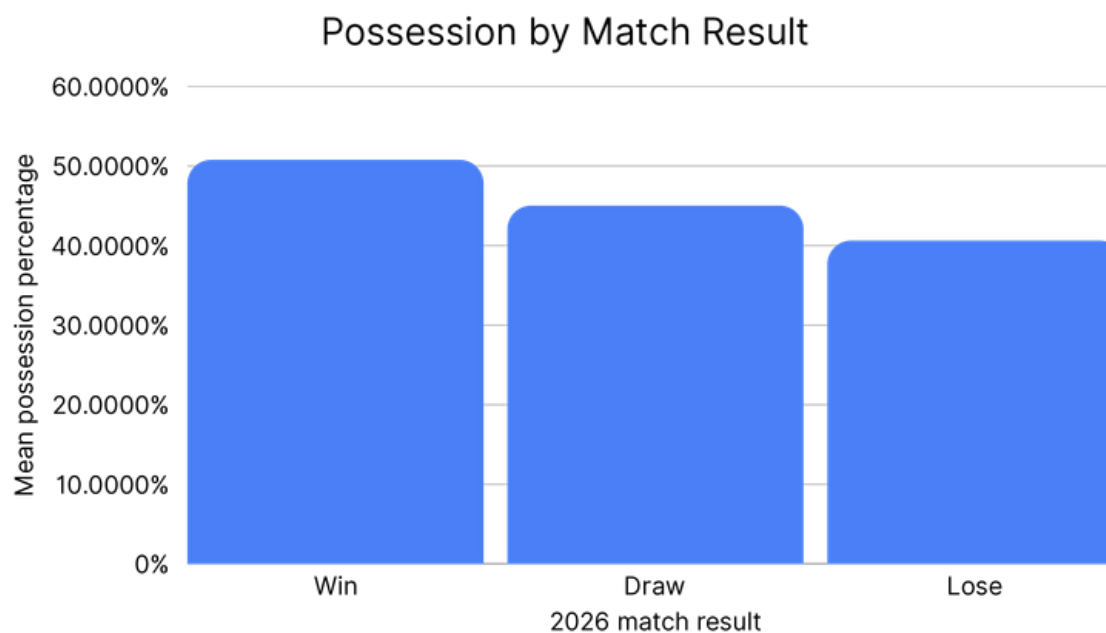
Figure 2. Selected average Men's World Cup goals per match over time.



Source: 04_sql_to_python_analysis.ipynb

Across 208 detailed 2026 team-match observations, mean possession was 50.7548% for wins, 44.9975% for draws, and 40.6095% for losses. This is a descriptive association, not a causal conclusion.

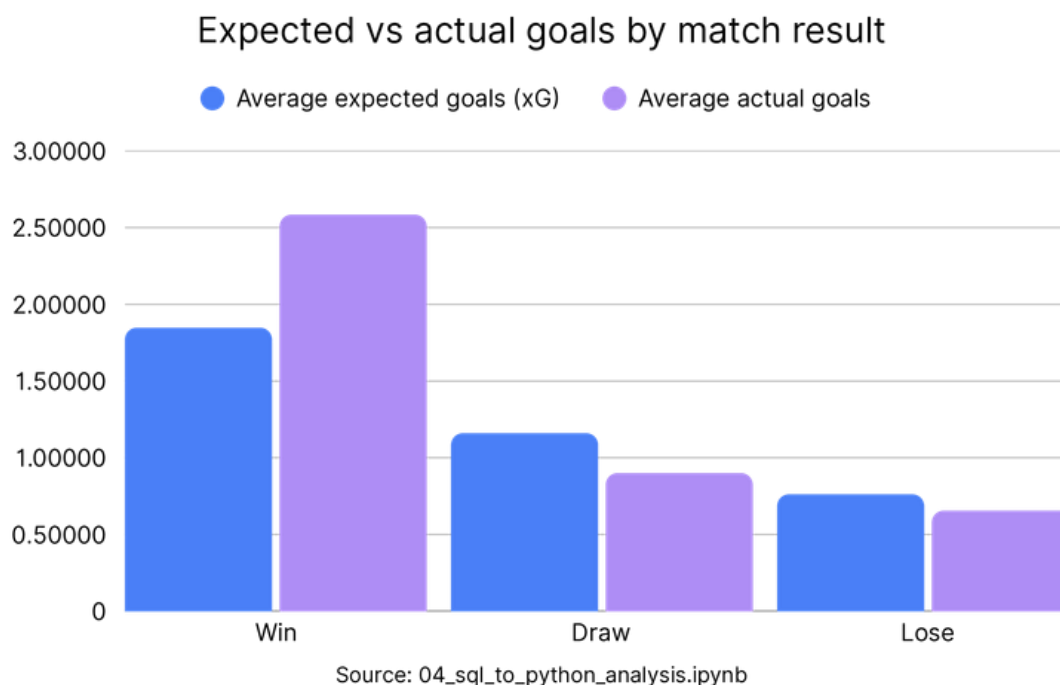
Figure 3. Mean possession percentage by detailed 2026 match result.



Source: 04_sql_to_python_analysis.ipynb

The xG comparison shows win observations at 1.846667 xG and 2.583333 actual goals; draw observations at 1.159750 xG and 0.900000 actual goals; and lose observations at 0.761667 xG and 0.654762 actual goals.

Figure 4. Average expected goals and actual goals by detailed 2026 result.

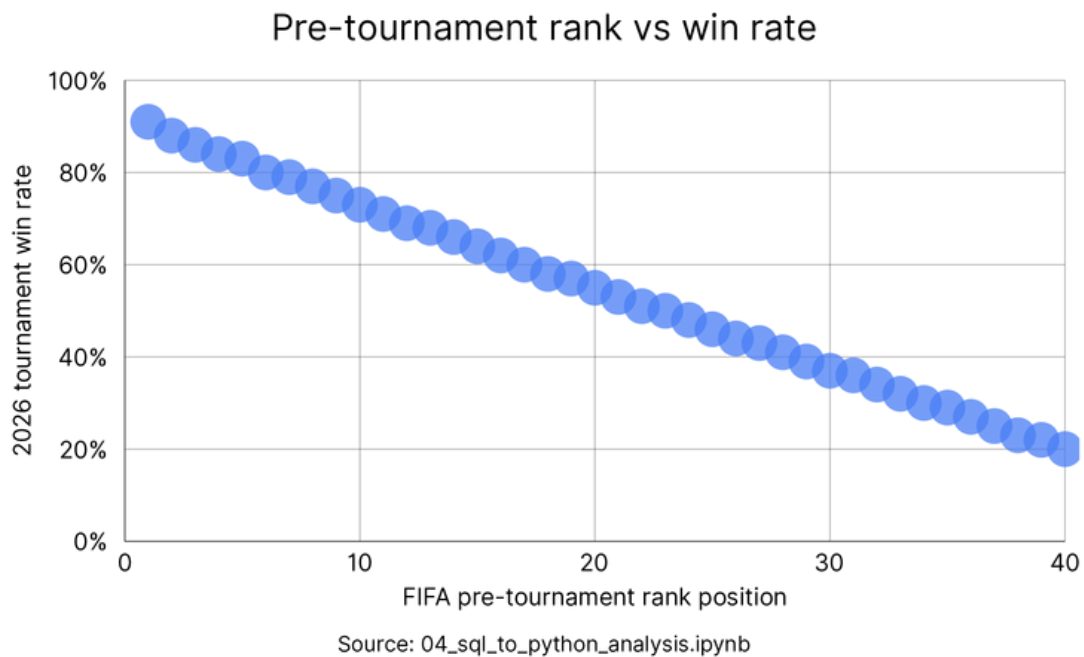


Mean pass completion was 87.142857% for wins, 83.475000% for draws, and 82.702380% for losses. Attendance-to-capacity ratios above 100% are treated as data-quality or historical-definition flags, not verified literal over-capacity utilization.

The FIFA pre-tournament ranking comparison uses 48 matched 2026 team observations. The Spearman result is $\rho = -0.7531$ with $p = 6.600889727e-10$. Because rank position 1 is best, the negative direction is consistent with better numerical rank being associated with higher win rate; it does not establish causation.

Figure 5. FIFA pre-tournament rank position versus 2026 win rate; individual point values are not supplied in the project documentation.

Interpretive boundary. Q22–Q25 and Q29 are descriptive analyses of the available detailed 2026 data. Derived measures and statistical associations should not be presented as causal findings.



7. Technology Stack & System Architecture

Technology	Documented role
Microsoft SQL Server	Relational database management and validated storage.
SQL	Schema creation, enrichment, validation, and 32 analytical queries.
Python	SQL-to-analysis integration, EDA, statistical checks, and visualization support.
Pandas and NumPy	DataFrame operations and numerical analysis.
Matplotlib and Seaborn	Analytical charts and distribution visualizations.
NetworkX	Passing-network visualization.
pyodbc	Connection from Python to SQL Server.
SciPy	Statistical analysis, including correlation checks.

Interactive website/frontend layer	Final presentation and user-experience layer for the documented results.
------------------------------------	--

Workflow: Data Sources → Data Collection → Data Cleaning → Data Integration → SQL Server Database → SQL Analytics → Python/Pandas/Statistics → Visualization → Interactive Website.

The uploaded text does not identify a frontend framework such as React or Vite. No additional frontend technology is therefore claimed here.

8. Conclusion & Future Improvements

8.1 Achieved system

The project delivers a validated 17-table FIFA World Cup analytics system containing 211,808 records. It combines relational database engineering, evidence-based enrichment, 32 analytical SQL queries, SQL-to-Python integration, statistical checks, 11 primary visualizations, 14 cross-query insights, and an interactive website presentation layer.

The documented workflow separates source facts from derived measures, statistical associations, and data-quality limitations. This distinction supports a defensible university-level project and makes the analytical conclusions appropriately bounded.

8.2 Limitations

- Attendance coverage is incomplete, and missing attendance is not zero.
- Historical capacities and attendance may use different periods or definitions.
- Raw totals are affected by tournament exposure and format changes.
- Detailed 2026 analyses have metric-specific source coverage limitations.
- The FIFA ranking result is an association within 48 matched observations, not a causal model.

8.3 Supported future improvements

- ☐ Extend the historical data coverage where reliable sources are available.
- ☐ Add real-time or more regularly refreshed data integration where supported.
- ☐ Develop more advanced predictive or machine-learning analysis.
- ☐ Expand dashboards and comparison views while preserving validated source semantics.
- ☐ Improve automated validation, filtering, and team or tournament comparison tools.

9. References / Project Artifacts

9.1 Documented source categories

- Fjelstul World Cup Database / DataHub.
- Detailed World Cup 2026 / FIFA Training Centre Data.
- FIFA World Ranking Data.
- FIFA Official 2026 Stadium Information.
- dim_players.csv

9.2 Named project artifacts

Artifact	Documented purpose
01_create_database.sql	Database creation.
02_bulk_insert_all.sql	Bulk loading and row-count validation.
03_update_missing_data.sql	Evidence-based enrichment, corrections, and verification.
04_analytical_queries.sql	Q1-Q32 analytical source of truth.
01_data_analysis.ipynb	Initial exploration and analysis preparation.
02_data_cleaning.ipynb	Cleaning and standardization.
03_data_validation.ipynb	Record-count, relationship, and source-coverage validation.
04_sql_to_python_analysis.ipynb	SQL Server integration, EDA, statistics, and visualizations.
FIFA_WorldCup_ERD.png	Referenced final ERD image; unavailable for direct inspection here.

Project artifacts: SQL scripts, Python notebooks, ERD, source datasets, and the interactive website are included as part of the project deliverables.